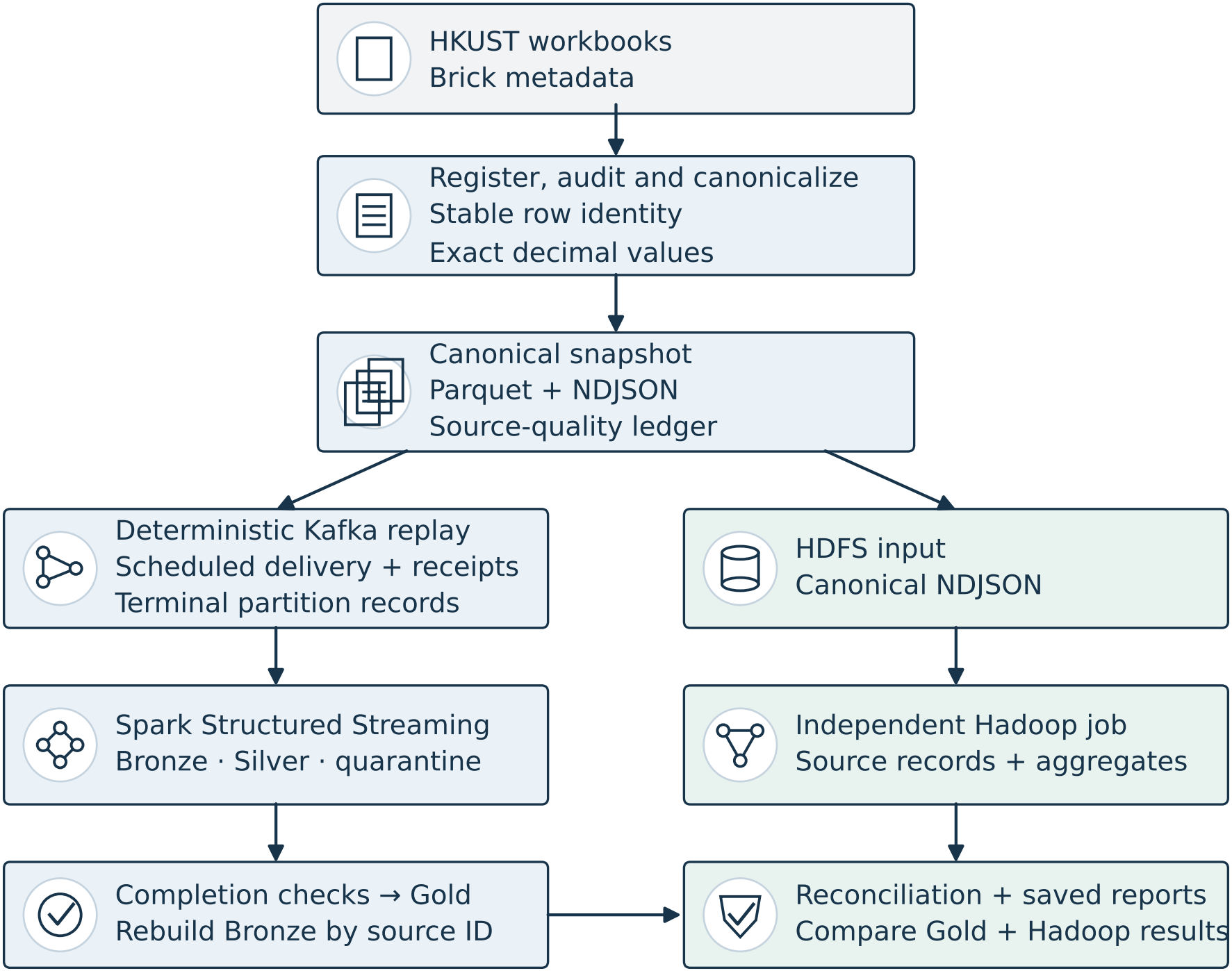
HKUST workbooks
Brick metadata
Register, audit and canonicalize
Stable row identity
Exact decimal values
Canonical snapshot
Parquet + NDJSON
Source-quality ledger
Deterministic Kafka replay
Scheduled delivery + receipts
Terminal partition records
HDFS input
Canonical NDJSON
Spark Structured Streaming
Bronze · Silver · quarantine
Independent Hadoop job
Source records + aggregates
Completion checks → Gold
Rebuild Bronze by source ID
Reconciliation + saved reports
Compare Gold + Hadoop results
Read-only dashboard: delivery accounting, comparison coverage and recorded costs